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1 · Constants, charge, vectors

☐ 1.1 · Electrostatic constants

$$k_e = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9, \quad \epsilon_0 = 8.85 \times 10^{-12}$$

When and how: Use k_e in Coulomb force, field, and potential; use ϵ_0 in Gauss law and vacuum capacitance. They encode the same interaction strength because $k_e = 1/(4\pi\epsilon_0)$.

Watch for: Do not substitute G for k_e . Convert charges to coulombs and lengths to metres first.

☐ 1.2 · Elementary charge, gravity, and charge counting

$$e = 1.602 \times 10^{-19} \text{ C}, \quad G = 6.674 \times 10^{-11}, \quad q = Ne$$

When and how: Use e to convert an electron count into net charge: N extra electrons gives $q = -Ne$ when N is a positive count. The sheet instead defines N as a signed integer. Use G for gravitational attraction.

Watch for: Neutral means equal positive and negative charge. It does not mean the object contains no charged particles.

☐ 1.3 · Conservation and vector reconstruction

$$Q_{\text{before}} = Q_{\text{after}}, \quad |\mathbf{E}| = \sqrt{E_x^2 + E_y^2}, \quad \theta = \text{atan2}(E_y, E_x)$$

When and how: Use charge conservation for transfer between objects in an isolated system. After adding electric-field components, use the square root to find magnitude and atan2 to find the correct direction quadrant.

Watch for: A zero vector total may result from cancellation of nonzero contributions. $\text{atan2}(E_y, E_x)$ is not simply $\arctan(E_y/E_x)$ in every quadrant.

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2 · Coulomb force, gravity & superposition

☐ 2.1 · Point-charge fields and test-charge force

$$\mathbf{R}_i = \mathbf{r} - \mathbf{r}_i, \quad \mathbf{E}(\mathbf{r}) = k_e \sum_i \frac{q_i \mathbf{R}_i}{|\mathbf{R}_i|^3}, \quad \mathbf{F} = q_0 \mathbf{E}$$

When and how: Given charges at coordinates, form each displacement $\mathbf{R}_i$ from its source to the observation point. Evaluate each vector contribution, add them, and multiply by q_0 if force is requested. This handles both 1D and 2D.

Watch for: $\mathbf{R}_i/|\mathbf{R}_i|^3$ includes direction and inverse-square magnitude. A negative source reverses its field; a negative test charge reverses its force relative to the total field.

☐ 2.2 · Two-body electric force and gravity

$$F_C = k_e \frac{|q_1 q_2|}{r^2}, \quad F_g = G \frac{m_1 m_2}{r^2}, \quad U_g = -\frac{G m_1 m_2}{r}$$

When and how: Use F_C for the magnitude between two point charges. Use F_g for gravitational attraction between masses and U_g for their gravitational potential energy with zero at infinity.

Watch for: The absolute value in F_C discards direction, so supply attraction or repulsion separately. U_g is negative for positive masses; force and energy have different powers of distance.

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☐ 3 · Continuous charge & trig substitution

☐ 3.1 · Choose a charge element

$$Q = \int dq, \quad dq = \lambda_q dl = \sigma dA = \rho_q dV, \quad \mathbf{E} = k_e \int \frac{\mathbf{R}}{R^2} dq$$

When and how: Use $\lambda_q dl$ for a wire, σdA for a sheet, or $\rho_q dV$ for a volume. Integrate dq for total charge, or integrate the vector $d\mathbf{E}$ for the field at a specified point.

Watch for: These are alternative descriptions of the source, not three densities to multiply together. Distance R and direction generally vary over the source.

☐ 3.2 · Field of a finite rod on its perpendicular bisector

$$E_y = k_e \lambda_q a \int_{-L}^L \frac{dx}{(a^2 + x^2)^{3/2}} = \frac{2k_e \lambda_q L}{a\sqrt{a^2 + L^2}}$$

When and how: Use this integral/result for a uniform rod from $-L$ to L , observed at $(0, a)$. Equal elements cancel horizontal components and reinforce vertical components; the projection supplies the extra factor a/R .

Watch for: Do not use this result at an arbitrary off-axis point. The denominator power $3/2$ includes both inverse-square distance and the component projection.

☐ 3.3 · Trigonometric substitution

$$x = a \tan \theta, \quad dx = a \sec^2 \theta d\theta, \quad R = a \sec \theta$$

When and how: When $a^2 + x^2$ occurs under a root, put $x = a \tan \theta$. Transform both dx and R , then replace the x bounds with angular bounds. This reduces the rod integral to a cosine integral.

Watch for: θ is the angle from the perpendicular through the rod midpoint. Forgetting the $\sec^2 \theta$ factor in dx gives the wrong integral.

☐ 3.4 · Evaluate and check the rod integral

$$E_y = \frac{k_e \lambda_q}{a} \int_{-\theta_0}^{\theta_0} \cos \theta d\theta, \quad \theta_0 = \tan^{-1}(L/a)$$

When and how: The endpoint angle is $\theta_0 = \arctan(L/a)$. Integrating cosine between $\pm\theta_0$ gives twice $\sin\theta_0$; $\sin\theta_0 = L/\sqrt{a^2 + L^2}$ restores the finite-rod result.

Watch for: For L much larger than a , recover an infinite line. For a much larger than L , recover a point charge with $Q = 2L\lambda_q$.

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4 · Flux, Gauss law & standard geometries

☐ 4.1 · Flux integral and Gauss law

$$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{A}, \quad \Phi_E = EA \cos \theta \text{ (uniform)}, \quad \oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

When and how: Use the open-surface integral for flux through any surface and $EA \cos \theta$ only when the field and angle are uniform. Use the closed integral to relate total outward flux to net enclosed charge.

Watch for: θ is measured from the surface normal. Zero net flux does not imply zero local field. An external charge contributes zero net closed-surface flux.

☐ 4.2 · Line, sheet, and spherical symmetry

$$E_{\text{line}} = \frac{\lambda_q}{2\pi\epsilon_0 r}, \quad E_{\text{sheet}} = \frac{\sigma}{2\epsilon_0}, \quad E_{\text{sphere}} = \frac{Q_{\text{enc}}(r)}{4\pi\epsilon_0 r^2}$$

When and how: Choose the line formula for an effectively infinite uniform line, the sheet formula for an infinite nonconducting sheet, and the spherical formula when the charge distribution is spherically symmetric.

Watch for: Gauss law always holds, but these direct field formulas require symmetry. A conducting surface has σ/ϵ_0 just outside, not $\sigma/(2\epsilon_0)$.

☐ 4.3 · Enclosed charge and uniform solid sphere

$$Q_{\text{enc}} = 4\pi \int_0^r \rho_q(s) s^2 ds; \quad \rho_q = \text{const} : E_{\text{in}} = \frac{\rho_q r}{3\epsilon_0}, \quad E_{\text{out}} = \frac{\rho_q R^3}{3\epsilon_0 r^2}$$

When and how: For a radially varying density, integrate concentric shells of volume $4\pi s^2 ds$ to the Gaussian radius. For constant density, use the linear interior field and inverse-square exterior field.

Watch for: Inside use only charge within r ; outside use the whole sphere of radius R . The two uniform-sphere formulas agree at $r = R$.

☐ 4.4 · Uniform solid cylinder

$$E_{\text{cyl,in}} = \frac{\rho_q r}{2\epsilon_0}, \quad E_{\text{cyl,out}} = \frac{\rho_q R^2}{2\epsilon_0 r}$$

When and how: Use the interior/exterior expressions for an infinitely long uniformly volume-charged cylinder of radius R . Cylindrical Gaussian area grows with r while enclosed interior charge grows with r^2 .

Watch for: A conducting cylinder is different: the field inside its metal is zero. For coaxial shells, compute the signed enclosed charge separately in each radial region.

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☐ 5 · Conductors, potential & work

☐ 5.1 · Conductor equilibrium

$$E_{\text{metal}} = 0, \quad E_t = 0, \quad E_{n,\text{out}} = \sigma / \epsilon_0$$

When and how: Use $E_{\text{metal}} = 0$ inside conducting material at electrostatic equilibrium. At its surface the tangential component is zero and the outward normal field just outside in vacuum is σ/ϵ_0 .

Watch for: A cavity containing charge can have a field even though the surrounding metal does not. The surface field sign follows the signed surface density.

☐ 5.2 · Potential from sources or a field

$$V = k_e \sum_i \frac{q_i}{R_i} = k_e \int \frac{dq}{R}, \quad \Delta V = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

When and how: For point charges or a continuous source, sum/integrate scalar potentials with zero at infinity where that reference is valid. If E is known instead, integrate $-\mathbf{E} \cdot d\mathbf{l}$ between two locations.

Watch for: Potential contributions add with charge signs but without vector components. For an infinite line, use a finite reference potential difference rather than zero at infinity.

☐ 5.3 · Potential slope, energy, and charge pairs

$$\mathbf{E} = -\nabla V, \quad E_x = -\frac{\partial V}{\partial x}, \quad U = qV, \quad U_{12} = \frac{k_e q_1 q_2}{r}$$

When and how: Use $\mathbf{E} = -\nabla V$ to get a field from a spatial potential function; use the partial derivative for one component. $U = qV$ uses the external potential at a test charge; U_{12} is pair interaction energy.

Watch for: Do not include a charge's own potential in qV . If summing pair energies for many charges, count each pair once. Field points toward decreasing V .

☐ 5.4 · Work and particle speed

$$W_E = -\Delta U = -q\Delta V, \quad K = \frac{1}{2}mv^2, \quad \Delta K = -q\Delta V$$

When and how: Use $W_E = -q\Delta V$ for electric work. If only electric work changes kinetic energy, set $\Delta K = -q\Delta V$ and use $K = \frac{1}{2}mv^2$ to find a speed or turning point.

Watch for: Work-energy gives speed, not the full velocity direction. If another force does work, include its contribution rather than using $\Delta K = -q\Delta V$ alone.

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6 · Useful axial fields & potential shapes

☐ 6.1 · Uniform ring on its axis

$$E_{\text{ring}}(z) = \frac{k_e Q z}{(z^2 + R^2)^{3/2}}, \quad V_{\text{ring}}(z) = \frac{k_e Q}{\sqrt{z^2 + R^2}}$$

When and how: Use these expressions for a ring of radius R and total charge Q, observed a signed axial distance z from its center. Transverse fields cancel; the axial field changes sign with z.

Watch for: At the center the field is zero but the potential is generally nonzero. Far from the ring both reduce to point-charge behavior.

☐ 6.2 · Uniform disk on its axis

$$E_{\text{disk}}(z) = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right) \quad (z > 0)$$

When and how: Use the disk field on the positive axis for uniform surface density σ . One derivation adds rings of charge, each contributing its axial field.

Watch for: This formula states $z > 0$. On the other side reverse the direction appropriately. A disk much wider than z approaches the infinite-sheet result.

☐ 6.3 · Force from an interaction-energy curve

$$F_r = -\frac{dU}{dr}; \quad U = \frac{C_{12}}{r^{12}} - \frac{C_6}{r^6} \Rightarrow F_r = \frac{12C_{12}}{r^{13}} - \frac{6C_6}{r^7}$$

When and how: Differentiate U with respect to outward separation r and change the sign. In the Lennard–Jones model, the short-range r^{-12} energy term creates repulsion while the negative r^{-6} term creates attraction farther out.

Watch for: Equilibrium is a zero slope, not zero energy. A local minimum is stable. C12 and C6 are material-specific constants, not electric charges.

☐ 6.4 · Arc center and solid-sphere potential

$$E_{\text{arc,center}} = \frac{2k_e \lambda_0 \sin \alpha}{R}, \quad V_{\text{solid,in}} = \frac{k_e Q}{2R} \left(3 - \frac{r^2}{R^2} \right)$$

When and how: For a uniform arc of total angle 2α , use the arc result at its center; symmetry leaves a field along the bisector. For a uniform solid sphere, use the potential result at $r < R$ with zero at infinity.

Watch for: The arc expression gives a signed bisector component for a chosen orientation; use an absolute value for magnitude. The solid-sphere result joins $k_e Q/R$ at the surface.

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7 · Capacitors & dielectric response

☐ 7.1 · Capacitance definition, plates, and combinations

$$C = \frac{Q}{\Delta V}, \quad C_{\text{plate}} = \frac{\epsilon A}{d}, \quad C_{\parallel} = \sum C_i, \quad \frac{1}{C_{\text{series}}} = \sum \frac{1}{C_i}$$

When and how: Use $C = Q/\Delta V$ for charge storage, $\epsilon A/d$ for large parallel plates, and the combination rules to simplify networks. Parallel elements share voltage; a simple isolated series chain has equal charge magnitude.

Watch for: C is geometry/material dependent, not proportional to Q when voltage changes on the same ideal capacitor. Use Q as the magnitude on either plate.

☐ 7.2 · Spherical and coaxial capacitors

$$C_{\text{sphere}} = 4\pi\epsilon \frac{ab}{b-a}, \quad C_{\text{coax}} = \frac{2\pi\epsilon L}{\ln(b/a)}$$

When and how: Use the spherical result for concentric conducting shells of radii a and b ; use the coaxial result for long coaxial cylinders of length L . Both assume the intervening dielectric is uniform.

Watch for: a is the smaller radius. Fringing must be negligible for the coaxial expression. Let b tend to infinity to get the capacitance $4\pi\epsilon a$ of an isolated sphere.

☐ 7.3 · Stored energy and energy density

$$U_C = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C} = \frac{1}{2}C(\Delta V)^2, \quad u_E = \frac{1}{2}\epsilon E^2$$

When and how: Integrate q/C as a capacitor is charged to Q to obtain its stored energy. Use the equivalent $Q^2/(2C)$ or $C(\Delta V)^2/2$ form depending on what is known; u_E is energy per unit volume.

Watch for: Decide what stays fixed: with a battery attached ΔV is fixed; disconnected Q is fixed. $u_E = \epsilon E^2/2$ assumes a linear, nondispersive dielectric in this electrostatic model.

☐ 7.4 · Polarization and electric displacement

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon \mathbf{E}, \quad \mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \quad \epsilon = \kappa \epsilon_0 = \epsilon_0 (1 + \chi_e)$$

When and how: Use $\mathbf{P}$ to describe dipole moment per volume and $\mathbf{D}$ to separate free charge from bound material charge. In a linear isotropic dielectric $\mathbf{P}$ is parallel to $\mathbf{E}$, allowing $\mathbf{D} = \epsilon \mathbf{E}$.

Watch for: $\mathbf{D} = \epsilon \mathbf{E}$ is a material relation, not a universal law. $\chi_e = \kappa - 1$ is dimensionless; $\mathbf{E}$ has units V/m, while $\mathbf{D}$ and $\mathbf{P}$ have units C/m².

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8 · Dipoles, mechanics & calculus starts

☐ 8.1 · Dipole rotation and orientation energy

$$\mathbf{p} = q\mathbf{d}, \quad \boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}, \quad U = -\mathbf{p} \cdot \mathbf{E}$$

When and how: Use $\mathbf{p} = q\mathbf{d}$ for separated equal and opposite charges, $\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$ for torque, and $U = -\mathbf{p} \cdot \mathbf{E}$ for orientation energy. A uniform field can give zero net force but a nonzero turning effect.

Watch for: $\mathbf{d}$ points from negative to positive charge. Parallel alignment minimizes energy; torque magnitude is $pE \sin\theta$ and is zero for parallel or antiparallel orientations.

☐ 8.2 · Distant dipole fields and potential

$$E_{\text{axial}} \simeq 2k_e p/r^3, \quad E_{\text{equator}} \simeq k_e p/r^3, \quad V \simeq k_e p \cos\theta/r^2$$

When and how: Use the axial or equatorial field magnitude far from a small dipole, and the potential expression at polar angle θ measured from $\mathbf{p}$. These approximate the difference between two nearly canceling point-charge sources.

Watch for: Require r much larger than the charge separation. The equatorial field points opposite $\mathbf{p}$; the axial field points along $\mathbf{p}$. At the equator $V = 0$ does not imply $E = 0$.

☐ 8.3 · Newton law and derivative definitions

$$\sum \mathbf{F} = m\mathbf{a}, \quad a = \frac{dv}{dt}, \quad v = \frac{dx}{dt}$$

When and how: Start dynamics with the vector sum of forces equal to $m\mathbf{a}$. Use $a = dv/dt$ and $v = dx/dt$ to move between acceleration, velocity, and position by differentiation or integration.

Watch for: Net force, not just one force, determines acceleration. Initial values are needed when integrating; derivatives of position alone cannot recover those constants.

☐ 8.4 · Constant-acceleration motion

$$a = \text{const} : v = v_i + at, \quad x = x_i + v_i t + \frac{1}{2}at^2, \quad v^2 = v_i^2 + 2a\Delta x$$

When and how: Use these kinematics formulas only when acceleration is constant along the chosen axis. Choose $v = v_i + at$ for time-to-speed, $x(t)$ for displacement with time, and $v^2 = v_i^2 + 2a\Delta x$ when time is absent.

Watch for: Apply each component separately in 2D. The squared-speed relation loses the velocity sign, so direction must come from the physical motion.

☐ 8.5 · Deflection between parallel plates

$$a_y = qE/m, \quad t = \ell/v_x, \quad y = \frac{1}{2}a_y t^2 \quad (v_{y,i} = 0)$$

When and how: For uniform transverse E , compute $a_y = qE/m$. If no horizontal force acts, transit time is plate length ℓ divided by v_x . With zero initial vertical speed and displacement, $y = \frac{1}{2}a_y t^2$.

Watch for: This gives displacement while inside the plates. Beyond them, motion is straight at the exit velocity if no other forces act. Include gravity if it is not negligible.

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9 · Oscillations, energy & damping

☐ 9.1 · Restoring force and natural period

$$F = -k_s x \Rightarrow \ddot{x} + \omega_0^2 x = 0, \quad \omega_0 = \sqrt{k_s/m}, \quad T = 2\pi/\omega_0$$

When and how: Use Hooke law to obtain the SHM equation and $\omega_0 = \sqrt{(k_s/m)}$. Period is $2\pi/\omega_0$. This identifies ideal mass-spring motion and supplies the starting point for a differential-equation derivation.

Watch for: x is measured from equilibrium. k_s is spring stiffness, not Coulomb's constant or wave number. Greater mass increases period; greater stiffness decreases it.

☐ 9.2 · SHM position, velocity, and acceleration

$$x = A \cos(\omega_0 t + \phi), \quad v = -A\omega_0 \sin(\omega_0 t + \phi), \quad a = -\omega_0^2 x$$

When and how: Use the sinusoidal solution for an undamped oscillator; differentiate to get v and a . At a turning point speed is zero and restoring acceleration has its greatest magnitude.

Watch for: ϕ sets the initial state. The velocity formula here corresponds to the displayed cosine displacement; do not change one phase convention without changing the derivatives.

☐ 9.3 · Energy at an oscillator position

$$U_s = \frac{1}{2}k_s x^2, \quad K = \frac{1}{2}k_s(A^2 - x^2), \quad E_{\text{tot}} = \frac{1}{2}k_s A^2$$

When and how: Use U_s for spring energy, K for kinetic energy at position x , and E_{tot} for total energy fixed by amplitude A . This often gives speed without first solving for time.

Watch for: These formulas assume ideal undamped SHM. x must lie between $-A$ and A ; K cannot be negative. At equilibrium energy is kinetic; at a turning point it is all spring energy.

☐ 9.4 · Pendulum period and driven-oscillator equation

$$T_{\text{pend}} = 2\pi\sqrt{\ell/g}, \quad m\ddot{x} + b\dot{x} + k_s x = F_0 \cos(\Omega t)$$

When and how: Use the pendulum period for small angular oscillations of a simple pendulum. Use the driven differential equation when damping proportional to velocity and an external sinusoidal force are present.

Watch for: Pendulum length ℓ runs from pivot to mass. Large-angle pendulum motion is not described by this amplitude-independent period. b is a damping coefficient, not a radius.

☐ 9.5 · Damped frequency and steady driven amplitude

$$\omega_d = \sqrt{\omega_0^2 - (b/2m)^2}, \quad A_{\text{drive}} = \frac{F_0}{\sqrt{(k_s - m\Omega^2)^2 + (b\Omega)^2}}$$

When and how: Use ω_d for free underdamped oscillations when $b/(2m) < \omega_0$. Use A_{drive} for the steady-state amplitude at forcing frequency Ω , after startup transients have decayed.

Watch for: Free damped motion and forced steady motion are different cases. With damping, the displacement-amplitude resonance is generally slightly below ω_0 ; at zero damping and exact resonance no finite steady amplitude exists.

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10 · Traveling wave: decode & differentiate

☐ 10.1 · Decode a sinusoidal traveling wave

$$y = A \cos \psi, \quad \psi = k_w x - \omega t + \phi, \quad k_w = \frac{2\pi}{\lambda}, \quad \omega = 2\pi f$$

When and how: Compare a supplied wave to $A \cos(kwx - \omega t + \phi)$. Read A, kw, ω , and ϕ directly; convert kw to wavelength and ω to frequency using 2π . Constant phase identifies direction.

Watch for: For positive kw and ω , minus ωt travels toward +x. A plus sign travels toward -x. Do not confuse angular frequency ω with cycles per second f.

☐ 10.2 · Period and propagation speed

$$T = \frac{1}{f} = \frac{2\pi}{\omega}, \quad v_w = \frac{\omega}{k_w} = f\lambda = \sqrt{T/\mu}$$

When and how: Use $T = 1/f$ for period and $v_w = \omega/k_w = f\lambda$ for waveform speed. For an ideal stretched string use $\sqrt{T/\mu}$ to connect speed to tension and mass per unit length.

Watch for: The speed relation $\sqrt{T/\mu}$ is string-specific. Source frequency can change while medium speed remains fixed; wavelength then adjusts.

☐ 10.3 · Motion of one point on the string

$$v_y = \partial_t y = A\omega \sin \psi, \quad a_y = \partial_t^2 y = -\omega^2 y$$

When and how: Hold x fixed and differentiate with respect to time to get transverse particle velocity and acceleration. Substitute the requested x and t only after taking the derivative.

Watch for: v_y is not wave speed v_w . The positive sine in v_y results from two minus signs: derivative of cosine and derivative of $-\omega t$.

☐ 10.4 · Verify the linear wave equation

$$\partial_x^2 y = -k_w^2 y \Rightarrow \boxed{\partial_t^2 y = v_w^2 \partial_x^2 y}$$

When and how: Differentiate twice in space and time. The same y appears with factors $-k_w^2$ and $-\omega^2$, yielding $\partial_t^2 y = v_w^2 \partial_x^2 y$ when $v_w = \omega/k_w$.

Watch for: This verifies the harmonic solution. General traveling shapes $f(x - v_w t)$ also satisfy the constant-speed wave equation; the PDE does not require a sinusoidal source.

☐ 10.5 · Derive the string PDE and calculate mean power

$$\mathcal{T} \partial_x^2 y dx = \mu dx \partial_t^2 y, \quad \overline{P} = \frac{1}{2} \mu \omega^2 A^2 v_w$$

When and how: For a short string element dx, small-slope tension forces give net transverse force $\mathcal{T} y_{xx} dx$ and mass μdx times acceleration. Cancel dx to derive the PDE. Use mean power for a sinusoidal traveling string wave.

Watch for: Assume uniform tension and density, small slopes, negligible loss. Doubling amplitude quadruples mean power at fixed frequency and medium.

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11 · Boundaries, interference & standing waves

☐ 11.1 · Phase difference and interference amplitude

$$\Delta\psi = k_w\Delta x - \omega\Delta t + \Delta\phi, \quad A_{\text{sum}} = 2A|\cos(\Delta\psi/2)|$$

When and how: Use the phase-difference expression to compare positions, times, or initial phases. For two equal-amplitude, same-frequency coherent waves, the cosine factor gives resultant amplitude.

Watch for: Constructive interference has phase difference $2\pi m$; destructive has $(2m+1)\pi$. The amplitude expression is not valid for arbitrary unequal amplitudes.

☐ 11.2 · Standing waves and two matching ends

$$y_1 + y_2 = 2A \sin(k_w x) \cos(\omega t), \quad f_n = \frac{nv_w}{2L} \quad (n = 1, 2, \dots)$$

When and how: Add equal opposite-traveling waves to get stationary nodes and antinodes. Use $f_n = nv_w/(2L)$ for a string fixed at both ends or a pipe open at both ends, neglecting end corrections.

Watch for: String fixed ends are displacement nodes. Open pipe ends are displacement antinodes and pressure nodes. Node spacing is $\lambda/2$; adjacent node-to-antinode distance is $\lambda/4$.

☐ 11.3 · One closed pipe end and beats

$$f_n = \frac{nv_w}{4L} \quad (n = 1, 3, 5, \dots), \quad f_{\text{beat}} = |f_1 - f_2|$$

When and how: Use odd- n quarter-wave modes for an open-closed pipe. Use the absolute frequency difference for the slow loudness modulation produced by nearby frequencies.

Watch for: The third harmonic of an open-closed pipe is $n = 3$; $n = 2$ is absent in the ideal model. Beat frequency is not the carrier pitch.

☐ 11.4 · Refraction and critical angle

$$n_1 \sin \theta_1 = n_2 \sin \theta_2, \quad n = c/v, \quad \sin \theta_c = n_2/n_1$$

When and how: Use Snell law for refraction at a stationary interface. With $n = c/v$, entering a higher-index medium bends toward the normal. Critical angle applies when crossing from higher to lower index.

Watch for: Measure both angles from the normal. Frequency stays continuous. For sound, use $\sin \theta_1/v_1 = \sin \theta_2/v_2$ rather than assuming optical indices; total internal reflection requires a valid critical-angle geometry.

☐ 11.5 · String-junction reflection and transmission

$$Z_j = \mathcal{T}/v_j, \quad r_A = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t_A = \frac{2Z_1}{Z_1 + Z_2}$$

When and how: Use these signed displacement-amplitude ratios at a junction of two ideal strings under the same tension. Incident motion travels from medium 1 toward medium 2; $Z = \mathcal{T}/v$ is the string impedance.

Watch for: A negative r_A means inversion. Amplitude ratios are not energy fractions: reflected power fraction is r_A^2 and transmitted is $(Z_2/Z_1)t_A^2$. A transmission amplitude greater than one need not violate energy conservation.

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12 · Sound, levels & Doppler

☐ 12.1 · Sound speed and intensity

$$v_s = \sqrt{B/\rho_m}, \quad I = \frac{P}{4\pi r^2} = \frac{p_{\text{rms}}^2}{\rho_m v_s} = \frac{1}{2} \rho_m v_s \omega^2 s_m^2$$

When and how: Use $\sqrt{B/\rho_m}$ for longitudinal sound in a fluid. Use $P/(4\pi r^2)$ for an isotropic source without absorption; use the pressure or displacement forms for a progressive sinusoidal sound wave.

Watch for: B here is bulk modulus, not magnetic field. The equality between pressure intensity and displacement intensity assumes the same medium and a traveling wave.

☐ 12.2 · Pressure amplitude, RMS, and level

$$p_m = \rho_m v_s \omega s_m, \quad p_{\text{rms}} = p_m / \sqrt{2}, \quad \beta = 10 \log_{10}(I/I_0)$$

When and how: Convert sinusoidal pressure amplitude to RMS by dividing by $\sqrt{2}$. Relate pressure amplitude to particle displacement through $p_m v_s \omega s_m$. Use β to express intensity relative to I_0 .

Watch for: Do not substitute peak pressure where RMS is required. I_0 is the conventional 10^{-12} W/m^2 reference; a negative dB level simply means below that reference.

☐ 12.3 · Compare two sound levels

$$\Delta\beta = 10 \log_{10}(I_2/I_1) = 20 \log_{10}(p_2/p_1)$$

When and how: For intensity ratios use $10 \log_{10}$; for pressure ratios in the same acoustic impedance use $20 \log_{10}$ because intensity is proportional to pressure squared.

Watch for: Pressure tripled means intensity multiplied by nine, and both calculations give +9.54 dB. The pressure-ratio equality needs the same medium/impedance.

☐ 12.4 · Doppler shift and induction reminder

$$f_o = f_s \frac{v_s + v_o}{v_s - v_{\text{src}}}, \quad \mathcal{E}_{\text{ind}} = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

When and how: Use Doppler for source/observer motion relative to the sound medium: positive v_o points toward the source, positive v_{src} toward the observer. The other two expressions describe magnetic flux and induced emf when flux changes.

Watch for: Approach raises observed frequency. The sound formula assumes classical, subsonic motion along the line of propagation. Faraday's minus sign expresses opposition to flux change; the displayed emf is for one loop (multiply by turns for identical linked flux).

My priority IDs: _____ Short reminder: _____